

Modeling of resistive plasma response in toroidal geometry using an asymptotic matching approach

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ABSTRACT

The method of solving the linear resistive plasma response, based on the asymptotic matching approach, is developed for full toroidal tokamaks by upgrading the resistive DCON code [A. H. Glasser, Z. R. Wang, and J.-K. Park, Phys. Plasmas **23**, 112506 (2016)]. The derived matching matrix, asymptotically matching the outer and inner regions, indicates that the applied three dimension (3-D) magnetic perturbations contribute additional small solutions at each resonant surface due to the toroidal coupling of poloidal modes. In contrast, the resonant harmonic only affects the corresponding resonant surface in the cylindrical plasma. The solution of the ideal outer region is critical to the asymptotic matching and is challenging in toroidal geometry due to the singular power series solution at the resonant surfaces. Thus, a systematic verification of the outer region Δ' matrix is made by reproducing the well-known analytical Δ' result in Furth *et al.* [Phys. Fluids **16**, 1054–1063 (1973)] and by making a quantitative benchmark with the PEST3 code [A. Pletzer and R. L. Dewar, J. Plasma Phys. **45**, 427–451 (1991)]. Finally, the reconstructed numerical solution of the resistive plasma response from the toroidal matching matrix is presented. Compared with the ideal plasma response, the global structure of the response can be affected by the small finite island at the resonant surfaces.

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I. INTRODUCTION

An axisymmetric tokamak plasma can be perturbed by small external three dimensional (3-D) magnetic perturbations that are applied through external coils. The modification of the plasma current and the equilibrium fields ($\vec{B}$) can lead to a non-axisymmetric state of plasma, which is the so-called 3-D plasma response. This phenomenon has been observed in experiments^{1–6} for years. Since the strength of the perturbed field is often several orders of magnitude smaller than that of the equilibrium field,^{7,8} the linear ideal magnetohydrodynamic (MHD) modeling has been extensively applied to understand the plasma response physics.^{9–14} The ideal plasma response studies greatly help to understand many experimental observations, e.g., the resonant magnetic perturbation^{15–20} and neoclassical toroidal viscosity (NTV) torque induced by the nonambipolar transport due to the presence of 3D perturbations.^{21–25} However, many experiments further indicate that non-ideal physics, such as the plasma resistive effect,^{26–36} can be important for determining the plasma response and affecting the aforementioned physics. In particular, one candidate mechanism for

edge-localized mode (ELM) suppression is caused by field penetration driving magnetic islands at the pedestal top due to the resistive plasma response.^{37,38} On the other hand, an experiment in the KSTAR tokamak also shows that the resistive plasma response plays an important role in determining the thermal electron induced NTV torque.³⁹ The resistive plasma physics closely relates to these important subjects. Therefore, developing a reliable resistive plasma simulation with great flexibility can greatly help to better understand and predict the resistive MHD behavior in tokamak experiments.

Different from the global resistive MHD simulation^{10,40–44} with high numerical cost, a resistive plasma response simulation, based on the asymptotic matching approach in full toroidal geometry,⁴⁵ is developed in this work. The approach, extensively applied to study tearing instabilities, has attractive features, such as high numerical efficiency and clear separation of the physics between the ideal outer and the resistive inner region. Following this approach, the plasma is partitioned into a set of ideal outer regions, which are connected through the resistive inner region located near each resonant surface with the

safety factor $q = m/n$. Here, m and n are the poloidal and toroidal wave numbers, respectively. The ideal outer region is governed by the zero-frequency ideal MHD equations⁴⁶ and does not need to solve the complicated resistive MHD model. The resistive inner region is a narrow area surrounding each rational surface, which has the flexibility to include different layer physics and plays a very important role in determining the global solution while solving either the tearing instability or the resistive plasma response. For instance, many studies^{47–54} indicate that the plasma rotation has a significant impact, such as the screening effect, on determining the resistive plasma response and island formation through the resistive inner layer. The asymptotic approach also enables us to recognize the impact of external perturbations in the resistive plasma response. In the present work, the dispersion relation in the matrix form, solving the resistive plasma response, is systematically developed in Sec. II. Since the most challenging part is to solve the outer region Δ' in the toroidal asymptotic matching approach, a systematic verification of Δ' is made in Sec. III to affirm the reliability of solving either tearing instability or the resistive plasma response in this approach. Finally, the numerical solution of the resistive plasma response is presented in Sec. IV.

II. RESISTIVE PLASMA RESPONSE MODEL IN THE TOROIDAL ASYMPTOTIC MATCHING APPROACH

A. Linearized resistive MHD

The linear modeling of the resistive plasma response, as well as the resistive instabilities, starts with the single fluid resistive MHD equations (1). Here, the formulation is described in the full toroidal geometry based on the Eulerian frame,

$$\gamma^2 \rho \vec{\xi} = -\nabla p + \vec{j} \times \vec{B} + \vec{j} \times \vec{b}, \quad (1a)$$

$$\gamma \vec{b} = \gamma \nabla \times (\vec{\xi} \times \vec{B}) - \nabla \times (\eta \vec{j}), \quad (1b)$$

$$p = -\vec{\xi} \cdot \nabla p - \Gamma P \nabla \cdot \vec{\xi}, \quad (1c)$$

$$\mu_0 \vec{j} = \nabla \times \vec{b}. \quad (1d)$$

Here, Eq. (1), with zero fluid equilibrium rotation, includes the perturbed quantities such as plasma displacement $\vec{\xi}$, perturbed plasma current $\vec{j}$, perturbed magnetic field $\vec{b}$, and perturbed pressure p , where $\vec{b}$ represents the vector quantity. The equilibrium quantities, such as plasma density, magnetic fields, current, and plasma pressure, are denoted as ρ , $\vec{B}$, $\vec{j}$, and P , respectively. η is the plasma resistivity. μ_0 is the vacuum permeability. The ratio of specific heats is $\Gamma = 5/3$. It is noted that γ is the eigenvalue in the problem of solving the eigenmode of resistive MHD instabilities. On the other hand, $\gamma = i2\pi f$ represents the frequency of the linear resistive plasma response while applying the external 3D magnetic perturbations with toroidal rotating frequency f . In the case of $\eta = 0$, Eq. (1) becomes the ideal MHD.

B. Galerkin method to solve the ideal outer region

The resistive DCON (RDCON) code⁴⁵ was developed to solve Eq. (5) and has been applied to understand tearing instability in the compression of spherical tokamak plasmas.⁵⁵ In this work, the RDCON code is upgraded to solve the resistive plasma response with zero plasma rotation. Following Ref. 45, this paper explains the singular Galerkin method and toroidal asymptotic matching process in detail. These details can help to better understand the way of the RDCON code solving the reliable outer region solutions and

theoretical basis of matching the outer and inner regions to solve the resistive plasma response. In the asymptotic matching, the ideal outer region is governed by the zero-frequency ideal MHD equations that are identical to the following Euler–Lagrange (E–L) equation (2). Therefore, the solution in the ideal outer region can be solved based on the E–L equation, which is derived from the energy minimization in the full toroidal geometry,⁴⁶

$$L\vec{\Xi} = -(\mathbf{F}\vec{\Xi}' + \mathbf{K}\vec{\Xi})' + (\mathbf{K}'\vec{\Xi}' + \mathbf{G}\vec{\Xi}) = 0. \quad (2)$$

In Eq. (2), the vector $\vec{\Xi} = \{\xi_\psi^m, m_{low} \leq m \leq m_{high}\}$ represents the M -vector of poloidal Fourier harmonics of perturbed radial displacement $\xi_\psi = \vec{\xi} \cdot \nabla \psi$ for a single toroidal mode with toroidal number n . ψ is the normalized poloidal flux. m_{low} and m_{high} are the lower and upper limits of poloidal mode number m . $M = m_{high} - m_{low} + 1$ is the total poloidal harmonic number in the computation. The coefficients $\mathbf{F}$, $\mathbf{K}$, and $\mathbf{G}$, in $M \times M$ matrix form, are derived based on the equilibrium quantities, where the detailed definitions of the quantities are described in Ref. 46. Similar to the Newcomb equation in a cylindrical geometry, Eq. (2) has the regular singular points at the resonant surfaces where $q(\psi) = m/n$, where the singularities can result in the resonant power series solutions of $\vec{\Xi}$ at the resonant surfaces. Near each side of resonant surfaces, there are the big power series solution $\vec{u}_{ip}^- \sim x^{-1/2-\sqrt{D_I}}$ and the small power series solution $\vec{u}_{ip}^+ \sim x^{-1/2+\sqrt{D_I}}$. Here, $x = |\psi - \psi_i'|$ is the distance to the resonant surface, ψ_i' is the value of ψ at the rational surface, and i is the index of the rational surface. $p = L, R$ represents the left and right sides of the rational surface. D_I is the Mercier index^{56–58} at ψ_i' . The small and big power series solutions $\vec{u}_{ip}^\pm$ of Eq. (2) have the form

$$\vec{u}_{ip}^\pm = x^{-1/2 \pm \sqrt{D_I}} \sum_{k=0}^{\infty} \vec{u}_{ip}^{(k)\pm} x^{k/2}, \quad (3)$$

where the details about the derivation of Eq. (3) and the technique of determining the coefficient $\vec{u}_{ip}^{(k)\pm}$ can be found in Sec. III of Ref. 46. It is noted that Eq. (3) includes both the integer power terms ($k/2 = 1, 2, 3, \dots$, when k is even) and the half integer power terms ($k/2 = 1/2, 3/2, 5/2, \dots$, when k is odd). Here, $\vec{u}_{ip}^{(k)\pm}$ with even k represents the contributions of resonant components to the power series solutions. When k is odd, $\vec{u}_{ip}^{(k)\pm}$ is derived from the coupling between the resonant and non-resonant components in the toroidal geometry. In the limit of cylindrical geometry, the half integer terms of Eq. (3) can vanish due to the decoupling between the resonant and non-resonant components, and Eq. (3) reproduces the conventional Frobenius expansions of big and small solutions with only the integer power terms. Moreover, besides the resonant solutions $\vec{u}_{ip}^\pm$, there are also the $M - 2$ non-resonant regular solutions in the toroidal geometry, which continuously cross the resonant surfaces. These solutions are denoted as $\vec{u}_{ip}^h$. Therefore, the solution of Eq. (2) can be expressed by the linear combination of resonant and non-resonant solutions as follows:

$$\vec{\Xi} = \sum_{i=1}^N \sum_{p=L,R} C_p^i \vec{u}_{ip}, \quad (4a)$$

$$\vec{u}_{ip} = \vec{u}_{ip}^- + \vec{u}_{ip}^s, \quad (4b)$$

$$\vec{u}_{ip}^s = \vec{u}_{ip}^h + \sum_{j=1}^N \sum_{q=L,R} \Delta_{pq}^{ij} \vec{u}_{jq}^+. \quad (4c)$$

Here, $\vec{u}$ is represented by the linear combination of $2N$ independent solutions $\vec{u}_{ip}$. N is the number of resonant surfaces in the plasma. As for each independent solution, the big solution $\vec{u}_{ip}^-$, as the driving term of Eq. (5), determines the $2N$ unknown Δ_{pq}^{ij} , as well as $\vec{u}_{ip}^h$, by solving the following inhomogeneous equation:

$$\mathbf{L}\vec{u}_{ip}^s = -\mathbf{L}\vec{u}_{ip}^- \quad (5)$$

In other words, the coefficient Δ_{pq}^{ij} , which is the essential term in the asymptotic matching process, represents the response of small solution $\vec{u}_{jq}^+$ to the big solution $\vec{u}_{ip}^-$ in the ideal outer region, where the response is illustrated in Fig. 1.

The RDCON code employs the finite element method (FEM), where the Hermite cubic basis functions are introduced in each element to represent $\vec{u}_{ip}^h$ since the solution, crossing the resonant surfaces, requires C^1 continuity. The basis functions in the elements are denoted as Φ_e , where e is the index of basis function. The quadratic form of the FEM is written as

$$\int \Phi_e^T \mathbf{L} \vec{u}_{ip}^s d\psi = - \int \Phi_e^T \mathbf{L} \vec{u}_{ip}^- d\psi. \quad (6)$$

Considering the nature of the small solutions, which are discontinuous at the resonant surface, two special elements are introduced in the neighborhood of each resonant surface. Figure 2 shows that there are two kinds of special elements and the normal elements at each side of the resonant surface. Each normal element (N element) contains four Hermite cubic polynomials, denoted as $\Phi_e = \{\phi_e^{00}, \phi_e^{01}, \phi_e^{10}, \phi_e^{11}\}$ as in Refs. 59 and 60. In the N elements, the solution can be expressed as $\vec{u}_{ip}^e = \vec{u}_{ip}^{0e} \phi_e^{00} + \vec{u}_{ip}^{1e} \phi_e^{01} + \vec{u}_{ip}^{0e} \phi_e^{10} + \vec{u}_{ip}^{1e} \phi_e^{11}$. The unknown coefficients $\vec{u}_{ip}^{0e}$ and $\vec{u}_{ip}^{1e}$ are the values of FEM solutions at the starting and ending points of the element interval, respectively. $\vec{u}_{ip}^{0e}$ and $\vec{u}_{ip}^{1e}$ are

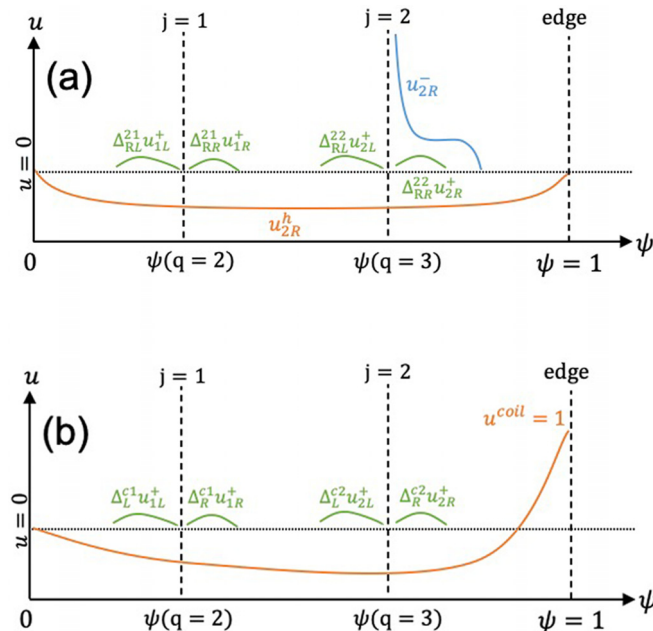


FIG. 1. Schematic plot of the small solution response to the big solution (a) and the external perturbation (b), respectively.

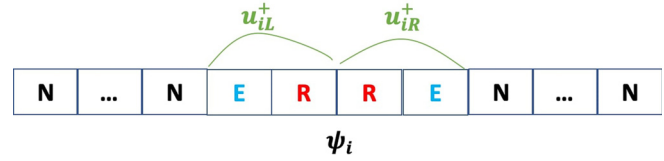


FIG. 2. The schematic plot of normal (N), resonant (R), and extension (E) elements around the resonant surface ψ_i .

their first order derivatives. In the special element, named resonant element (R element), besides the Hermite basis, the basis function Φ_e further includes the small resonant solution $\vec{u}_{jq}^+$ as the additional basis function. The other special element, named extension element (E element), contains the Hermite basis and the additional basis function $\vec{e}_{jq}$, which plays a role in making the value, and the first order derivative of $\vec{u}_{jq}^+$ vanishes smoothly while connecting the R element and N element. $\vec{e}_{jq}$ is developed from the Hermite cubic basis functions. For instance, in the E element on the right hand side of the resonant surface, the addition basis is denoted as $\vec{e}_{jq} = \vec{u}_{jq}^+ \phi_e^{00} + \vec{u}_{jq}^+ \phi_e^{01}$, where $\vec{u}_{jq}^+$ and $\vec{u}_{jq}^+$ are the values of small solution and its first order derivative at the grid point connecting the R and E elements. Finally, in different elements, $\vec{u}_{ip}^s$ can be represented as

$$\vec{u}_{ip}^s = \begin{cases} \vec{u}_{ip}^e, & \text{N element} \\ \vec{u}_{ip}^e + \Delta_{pq}^{ij} \vec{u}_{jq}^+, & \text{R element} \\ \vec{u}_{ip}^e + \Delta_{pq}^{ij} \vec{e}_{jq}, & \text{E element} \end{cases} \quad (7)$$

In Eq. (7), the unknown coefficients, $\vec{u}_{ip}^{0e}$, $\vec{u}_{ip}^{1e}$, $\vec{u}_{ip}^{0e}$, $\vec{u}_{ip}^{1e}$, and Δ_{pq}^{ij} , can be solved by substituting this equation into Eq. (6). It is noted that $\vec{u}_{jq}^+$ in the R element and $\vec{e}_{jq}$ in the E element share the same Δ_{pq}^{ij} .

In the RDCON code, besides the aforementioned numerical scheme, grid packing around each resonant surface is applied to ensure that the numerical solution is well converged. On the right hand side of Eq. (5), the big solution $\mathbf{L}\vec{u}_{ip}^-$, as the driving term, is also the asymptotic solution at the resonant surface. As illustrated in Fig. 1(a), it is similar to the treatment of the small solution in R and E elements in that $\mathbf{L}\vec{u}_{ip}^-$ is smoothly truncated, while it is away from the resonant surface. In particular, the solution of Δ_{pq}^{ij} is unique, no matter where the truncation of $\mathbf{L}\vec{u}_{ip}^-$ is. The uniqueness of the solution is described in Ref. 61. The RDCON simulation also indicates that Δ_{pq}^{ij} is not sensitive to the truncation of $\mathbf{L}\vec{u}_{ip}^-$. Moreover, the boundary condition of solving Eq. (5) is $\vec{u}_{ip}^s = 0$ at the magnetic axis $\psi = 0$. At the plasma boundary with $\psi = 1$, different boundary conditions can be imposed. For instance, the fixed boundary problem can be solved to get Δ_{pq}^{ij} with given $\vec{u}_{ip}^s$ at $\psi = 1$. Imposing the vacuum response matrix can solve the free boundary problem with the vacuum region.⁴⁵

C. Matching outer and inner regions to solve the resistive plasma response

Solving Eq. (6) can determine $2N$ independent solutions of $\vec{u}_{ip}$ in Eq. (4a). In order to get the full solution from these independent solutions, it is essential to match the outer and inner regions asymptotically to determine the coefficients C_p . Here, the Glasser, Greene, and

Johnson (GGJ) model,⁵⁸ as an inner region model, is considered. The model is developed from Eq. (1) in toroidal geometry. In the outer region, the big and small solutions of Eq. (3) are ascending power series. In contrast, the GGJ model has two descending power series with the general form shown by Eq. (8). Since the GGJ model is a six order ordinary differential equation (ODE), there are other two exponential big and two exponential small solutions. Solving the GGJ model and matching with the outer region require excluding the four exponential solutions,^{45,58}

$$\bar{\mathbf{u}}_{jq}^{\pm} = X^{-1/2 \pm \sqrt{-D_I}} \sum_{k=0}^{\infty} \bar{\mathbf{u}}_{jq}^{(k)\pm} X^{-2k}. \quad (8)$$

It is noted that Eq. (8) is a scalar function since the GGJ model considers only the resonant harmonic and ignores the other non-resonant harmonics through a sophisticated ordering assumption.⁵⁸ In the equation, the independent variable $X = |\psi - \psi_i^r|/X_0$ is the normalized distance to the resonance surface ψ_i^r . The scale factor X_0 , estimating the inner layer width, is defined in the appendix of Ref. 45. Similar to the outer region for $x \rightarrow 0$, while $X \rightarrow \infty$, there are two independent solutions on the left and right sides of the inner region, where the solutions have the form as shown by the following equation:

$$\bar{\mathbf{u}}_{jq} = \bar{\mathbf{u}}_{jq}^- + \bar{\Delta}_{qL}^j \bar{\mathbf{u}}_{jL}^+ + \bar{\Delta}_{qR}^j \bar{\mathbf{u}}_{jR}^+, \quad (9)$$

j is the index of the resonant surface. $q = L, R$ represents the left and right sides of resonant surface j . The various methods of solving the GGJ model to determine the unknown inner region $\bar{\Delta}$ value have been established in Refs. 45 and 62. Here, the DELTAC code, developed in Ref. 45, is employed to solve the GGJ model. It is noted that the DELTAC code, adopting a similar numerical treatment as the RDCON code, uses the big solution to drive the response of the small solutions to determine $\bar{\Delta}_+$ and $\bar{\Delta}_-$, as well as $\bar{\Delta}_{qp}^j$ in the inner region. $\bar{\Delta}_+$ and $\bar{\Delta}_-$ represent the conventional even and odd parities of inner region solutions.⁵⁸

To solve the resistive plasma response, there are two factors affecting the small solutions in the outer region. One factor is the small solution driven by the big solution $\bar{\mathbf{u}}_{ip}^-$, which can be obtained by solving Eq. (5) with the boundary condition $\bar{\mathbf{u}}_{ip}^- = 0$ at $\psi = 1$. Since there are N resonant surfaces, the $2N$ big solution $\bar{\mathbf{u}}_{ip}^-$ can drive the $2N$ independent solution of $\bar{\mathbf{u}}_{ip}^s$ in Eq. (4b) and consequently determine the coefficient of $\bar{\Delta}_{pq}^{ij}$ in Eq. (4c). The other factor is the response of small solutions to the applied external 3D perturbations, which provides one more independent solution and contributes the additional $\bar{\Delta}_{jq}^c$ at each resonant surface to match with the inner region. Figure 1(b) illustrates the response small solution in the presence of an edge perturbation. Here, $\bar{\Delta}_{jq}^c$ can be determined by simply solving the following fixed boundary problem:

$$\mathbf{L}\bar{\mathbf{u}}_{ip}^s = 0. \quad (10)$$

The boundary condition is the given fixed edge perturbation $\bar{\mathbf{u}}_{ip}$ at $\psi = 1$. Similar to the determination of the tearing stability through the matching condition,⁴⁵ the resistive plasma response, as the result of the linear combination of $2N + 1$ independent solutions, can be determined by matching these outer region solutions with the resistive inner layers through the asymptotic matching process.

It is noted that Eq. (3) in the outer region is an ascending power series and Eq. (8) in the inner region is a descending power series. The

asymptotic matching requires that the matching happens in an overlapping region near the resonant surface where the zeroth order term is dominant in both Eqs. (3) and (8), and their high order terms vanish while $x \rightarrow 0$ and $X \rightarrow \infty$, respectively. The zeroth order terms in Eqs. (3) and (8) can be match on each side of every resonant surface. The toroidal matching conditions of the resistive plasma response, based on the fundamental definition of asymptotic matching, are developed,

$$C_L^j - c_L^j = 0, \quad (11a)$$

$$C_R^j - c_R^j = 0, \quad (11b)$$

$$\sum_{i=1}^N \sum_{p=L,R} C_p^i \Delta_{pL}^{ij} - \left[c_L^j \bar{\Delta}_{LL}^j(f) + c_R^j \bar{\Delta}_{RL}^j(f) \right] = -\Delta_L^j, \quad (11c)$$

$$\sum_{i=1}^N \sum_{p=L,R} C_p^i \Delta_{pR}^{ij} - \left[c_L^j \bar{\Delta}_{LR}^j(f) + c_R^j \bar{\Delta}_{RR}^j(f) \right] = -\Delta_R^j. \quad (11d)$$

Equations (11a) and (11b) represent the matching of big solutions between the inner and outer regions at the left and right of each resonant surface, respectively. Similarly, the small solutions are matched at each side of resonant surfaces through Eqs. (11c) and (11d). On the other hand, the eigenvalue γ in the GGJ model⁵⁸ needs to be redefined as the rotating frequency of applied external fields, $\gamma = i2\pi f$, in the linear plasma response calculation. Therefore, the inner region $\bar{\Delta}_{qp}^j$ is the function of f in Eq. (11). Since the resistive MHD equation (1) becomes singular when $\gamma = 0$, the rotating perturbation with the finite f is required in the present GGJ model. In order to solve the resistive response with the static perturbation, the GGJ model can be extended to include the plasma fluid rotation as the Doppler shift term in γ . The effect of plasma rotation on the resistive response will be investigated in the future work.

In Eq. (11), Δ_p^j is contributed by the small solution at the p side of the j th resonant surface, which is driven by the external perturbations. C_p^j and c_p^j are the unknown coefficients of the big solution in the outer and inner regions, respectively. The coefficients need to be determined by Eq. (11). The general form of Eq. (11) can be written as $(\bar{\Delta}_{out} - \bar{\Delta}_{in}(i2\pi f))\mathbf{C} = -\Delta_{coil}$, where $\bar{\Delta}_{out}$ and $\bar{\Delta}_{in}(i2\pi f)$ represent the outer and inner regions' $\bar{\Delta}$ matrices, respectively. Δ_{coil} is the contribution of small solutions induced by the external perturbations. Specifically, this general form is consistent with the result in section 7 of Ref. 33 in the cylindrical geometry. Note $\bar{\Delta}_{in}$ in Eq. (11) has been converted to the same dimension as $\bar{\Delta}_{out}$ by multiplying $X_0^{-2\sqrt{-D_I}}$ on the original $\bar{\Delta}_{in}^j$.⁶² Consequently, the resistive plasma response with fixed boundary conditions can be determined by $\mathbf{C} = -(\bar{\Delta}_{out} - \bar{\Delta}_{in}(i2\pi f))^{-1} \Delta_{coil}$ to finally solve C_p^j and c_p^j .

In the code development, the simulation module of 3D external coils is provided by the IPEC code,⁹ where the IPEC code is originally developed to solve the ideal plasma response. In order to couple the RDCON and DELTA codes to the IPEC code, the resistive plasma response is solved with fixed boundary conditions $\xi_{\psi}^{m=m'} = 1$ and $\xi_{\psi}^{m \neq m'} = 0$ at $\psi = 1$, which is the boundary connecting plasma to the vacuum. Once these M independent resistive solutions are determined, the connection between the fixed boundary solutions and vacuum solution with the external coils can be made by the permeability matrix that IPEC provides through the virtual casing principle.⁹ The virtual casing calculates the surface current equivalent to the currents

in the external coils by taking the difference in the tangential fields just outside and inside the plasma-vacuum boundary given a normal field distribution. The relationship between a tangential field and a normal field remains the same for the resistive solutions because of the same E–L equation (2) in the outer region, which is valid as long as the plasma vacuum boundary is not too close to a resonant surface. This procedure, solving the linear combination of M independent solutions in the IPEC code, results in the final solution of the resistive plasma response.

III. VERIFICATION OF OUTER REGION SOLUTION SOLVED BY THE RESISTIVE DCON CODE

Solving the reliable outer region solution, such as Δ_{out} matrix, is considered to be challenging and critical to determining the full resistive solutions in the toroidal asymptotic matching approach. Though a benchmark of outer region Δ' between the RDCON code and the MARS code has been made,⁴⁵ the benchmark is based on the simplified equilibrium with a large aspect ratio and only the $q=2$ resonant surface included. Meanwhile, the MARS code, as a global MHD code, cannot calculate the Δ' value directly, which is inferred from the inner region based on the GGJ model in the cylindrical limit.⁶³ In this section, the outer region solution, solved by the RDCON code, is verified by reproducing the well-known analytical Δ' results at $q=2$ and $q=3$ surfaces.⁶⁴ Furthermore, the benchmark of Δ' between the RDCON code and the PEST3 code is made by employing the strongly shaped equilibrium with a low aspect ratio and multiple resonant surfaces, which is close to the experimental case. Here, the PEST3 code^{61,65} is the other one having the capability of evaluating the Δ' matrix in the full toroidal geometry. This systematic benchmark eventually confirms that the RDCON code can solve the reliable outer region solution. In addition, the STRIDE code,⁶⁶ targeting the real time MHD stability computation, is developed based on the DCON code and solves exactly the same E–L equation (2) by utilizing the multiple shooting method. Following the RDCON code, the STRIDE code is also able to solve the same outer region Δ' , which agrees well with the result of the RDCON code.⁶⁶ The benchmark in this work can also inform the reliability of the STRIDE code solving the outer region Δ' .

For the convenience of comparing with existing theory and results, the conventional tearing index, outer region Δ' , is used for validating the RDCON solution. Since the resonant surfaces are coupled to each other in the full toroidal geometry, it results in the $\Delta' = (\Delta'_{ij})_{i \times j}$ matrix form, which can be evaluated by Δ'_{pq} using the following expression:⁴⁵

$$\Delta'_{ij} = \Delta'_{RR} - \Delta'_{LR} - \Delta'_{RL} + \Delta'_{LL}. \quad (12)$$

Here, Δ'_{ij} is due to the coupling of the i th resonant surface to the j th one. In the cylindrical limit, where the aspect ratio $A = R/a \rightarrow \infty$, $\Delta'_{ij} \rightarrow 0$ when $i \neq j$. a is the minor radius and R is the major radius. As for the case $i=j$, $\Delta'_{ii} = \Delta'_{RR} + \Delta'_{LL}$ recovers the cylindrical results of the outer region Δ' . The vanishing of Δ'_{RR} , Δ'_{LL} ($i \neq j$) and the off diagonal terms in the Δ' matrix are due to the decoupling between the resonant and non-resonant components. First, the RDCON code is validated by reproducing the well-known cylindrical Δ' results that are solved by Furth *et al.*⁶⁴ analytically. Since the RDCON code cannot directly solve Δ' with a cylindrical equilibrium, the Corsica code⁶⁷ is used to generate a series of toroidal equilibria that have a large aspect ratio $A=10$, a circular cross section, and low pressure, toroidal

$\beta \sim 3.408 \times 10^{-4} < 10^{-3}$. In addition, the same current profile as the peaked model in Ref. 64 is used in the equilibrium generation. These equilibria are stable to the ideal kink mode. The dimensionless quantity $r_0 \Delta'$ for the $n=1$ mode is solved by the RDCON code. The locations of the $q=2$ and $q=3$ resonant surfaces are scanned separately. The results of $r_0 \Delta'$ for $(m, n) = (2, 1)$ and $(3, 1)$ harmonics are presented in Fig. 3, where r_0 and r_s refer to the minor radius a and the location of the resonant surface, as defined in Ref. 64. There are two cases of scans with different wall locations $x_b = 1.35r_0$ and $x_b = 6r_0$. Generally, Fig. 3 well reproduces the behavior of $r_0 \Delta'$ in Fig. 1 of Ref. 64, where the values of $r_0 \Delta'$ for both $(2, 1)$ and $(3, 1)$ modes are decreased when the resonant surface moves toward the plasma edge. Particularly, the values of $r_0 \Delta'$ agree with the results of Ref. 64 quantitatively, when the resonant surface is getting close to the core of the plasma. The $r_0 \Delta'$ values of $(2, 1)$ mode are less positive (more stable) when the ideal wall is close to the plasma boundary, which also agrees with the cylindrical results. However, it is noted the $r_0 \Delta'$ values of $(2, 1)$ and $(3, 1)$ modes are less sensitive to the change in the wall location than that in Fig. 1 of Ref. 64. This slight difference can be due to the factors, e.g., toroidal geometry and the finite small pressure in the equilibrium reconstruction. Moreover, as expected, the off diagonal terms of the Δ' matrix are much smaller than the diagonal terms in this large aspect ratio plasma since the coupling effect between the resonant and non-resonant harmonics is greatly decreased while $A \rightarrow \infty$.

On the other hand, the GGJ model has shown that the pressure can play an important role in stabilizing the tearing instability from the inner region.⁵⁸ Here, the pressure effect in the outer region is also investigated by the RDCON code. Figure 4 shows that the plasma pressure is increased in the scan of the resonant surface, which finds that the $r_0 \Delta'$ values at both $q=2$ and $q=3$ surfaces are amplified and more positive due to the increase in plasma pressure. In contrast to the stabilizing effect of pressure in the GGJ inner region, the more positive Δ' value, which is proportional to the magnetic driving energy of tearing mode,⁶⁴ implies more free energy in the outer region to drive the tearing instability. The different impact implies that the matching

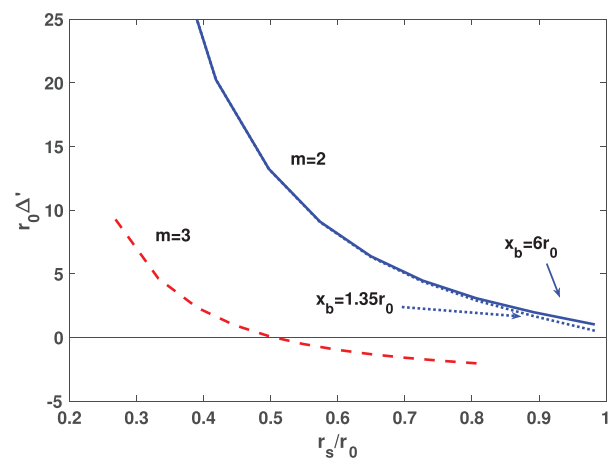


FIG. 3. The dependence of $r_0 \Delta'$ for $n=1$ mode on the location of the resonant surface r_s/r_0 with two different wall locations, $x_b = 1.35r_0$ and $x_b = 6r_0$, which reproduces the results in Ref. 64.

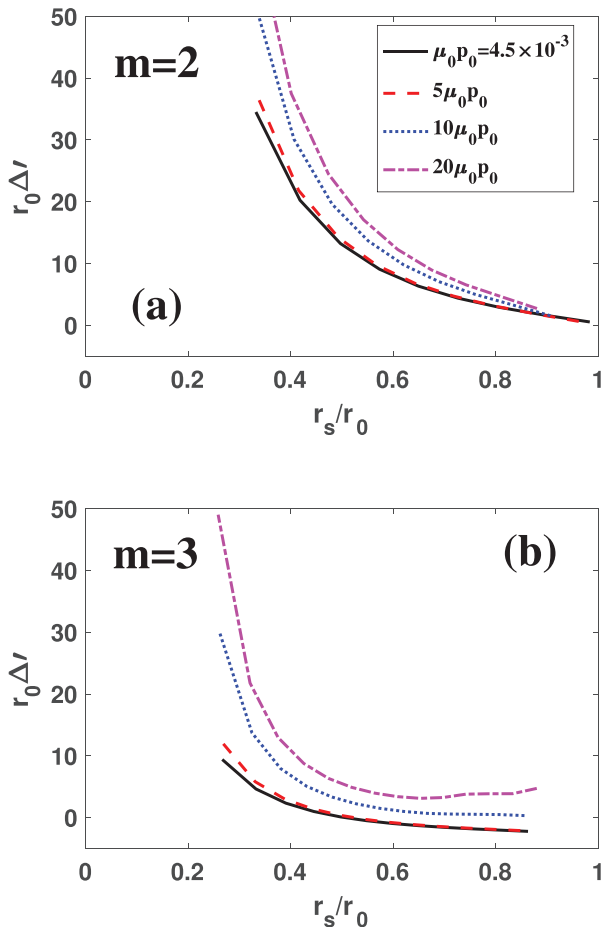


FIG. 4. The dependence of $r_0 \Delta'$ for $n=1$ mode on the location of the resonant surface r_s/r_0 with different plasma pressures, where the highest pressure is 20 times higher than original one. (a) and (b) correspond to Δ' at $q=2$ and $q=3$ resonant surfaces, respectively.

between the outer and inner regions to find the mode eigenvalue is critical to finally determine the tearing mode stability.

The PEST3 code evaluates the Δ' matrix by using the integral representation in the full toroidal geometry.^{61,65} In contrast, the RDCON code directly extracts the Δ matrix from the coefficient of small solutions, which is implemented as the extra basis function in the resonant element. Here, the Δ' matrix can be converted from the Δ matrix based on Eq. (12). A benchmark of Δ' values between RDCON and PEST3 codes is made by scanning β_N with a highly shaped plasma, where the plasma shape, q profile, and pressure profile are shown in Fig. 5. Both codes use the equilibria generated by the Corsica code to evaluate the dimensionless quantity, $\psi^{2\mu} \Delta'$. The diagonal terms of the Δ' matrix at different surfaces are compared, where ψ is the corresponding resonant surface and $\mu = -1/2 - \sqrt{D_I}$. Figures 6(a) and 6(b) present the benchmark of diagonal terms Δ' at $q=2$ and $q=3$, respectively. The two codes show very good quantitative agreement at low β_N and similar behavior of Δ' amplification while approaching the no-wall β limit, $\beta_N 3.5$. Moreover, a discrepancy of

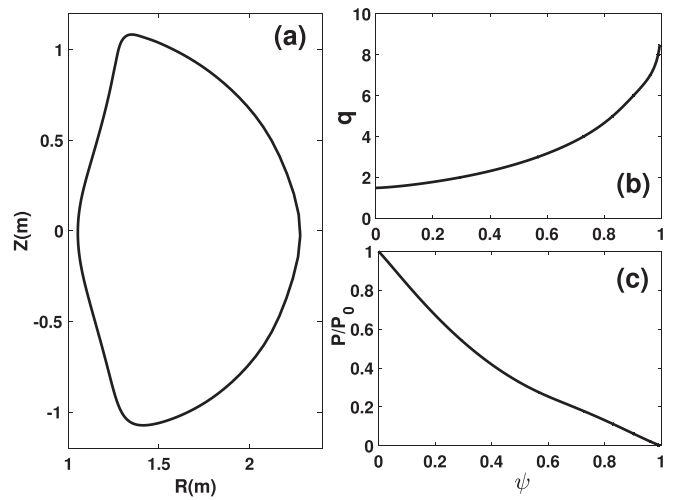


FIG. 5. (a) Cross section of simulated D-shape equilibrium. (b) and (c) present the corresponding safety factor (q) and equilibrium pressure.

the Δ' value between the two codes is expected while approaching the no-wall β limit since PEST3 only keeps the first order term in the big and small solutions and RDCON keeps arbitrary high order terms in the power series. Though the high order terms in Eq. (3) are not considered in the asymptotic matching process, these terms are still essential to keep the convergence of outer region solutions, while solving Eq. (5). Particularly, the big solution is getting more singular, while ∇P is increased. The high order terms play an important role in reducing the numerical residue while solving Eq. (3). In this benchmark, the RDCON code is set as $k=6$ in Eq. (3) to solve the well converged Δ' matrix.

In the cylindrical case, $\Delta' \rightarrow \infty$, while the plasma reaches the marginal stability of ideal kink mode. Here, both codes show a singular behavior of the Δ' value, where the Δ' value of the PEST3 code is more singular than that of the RDCON code. In principle, RDCON has better convergence of Δ' than the PEST3 code, as aforementioned. The less singular behavior of RDCON results may be due to the complexity of toroidal effects, which will require further effort to get better understanding in the future. Moreover, the growth rate, solved through the toroidal asymptotic matching approach by using the RDCON and DELTAC codes, has also been compared with the MARS-F code⁴⁵ which directly solves the global resistive MHD as an eigenvalue problem. All benchmark and comparison confirm the reliability of the newly developed toroidal asymptotic matching approach to solve the resistive plasma response.

IV. NUMERICAL SOLUTION OF THE RESISTIVE PLASMA RESPONSE

Finally, the resistive plasma response, based on the asymptotic matching condition (11), is illustrated. In this section, as described in Sec. II C, the simulation of the resistive plasma response is performed by coupling the RDCON and DELTAC codes to the IPEC codes. The ideal response is computed by the IPEC code. In the computation, the same equilibrium, as shown in Fig. 5, is adopted, where the uniform Lundquist number $S = 10^7$ is assumed in the resistive plasma. The very slow rotating magnetic perturbation with $f = 1$ Hz is applied in

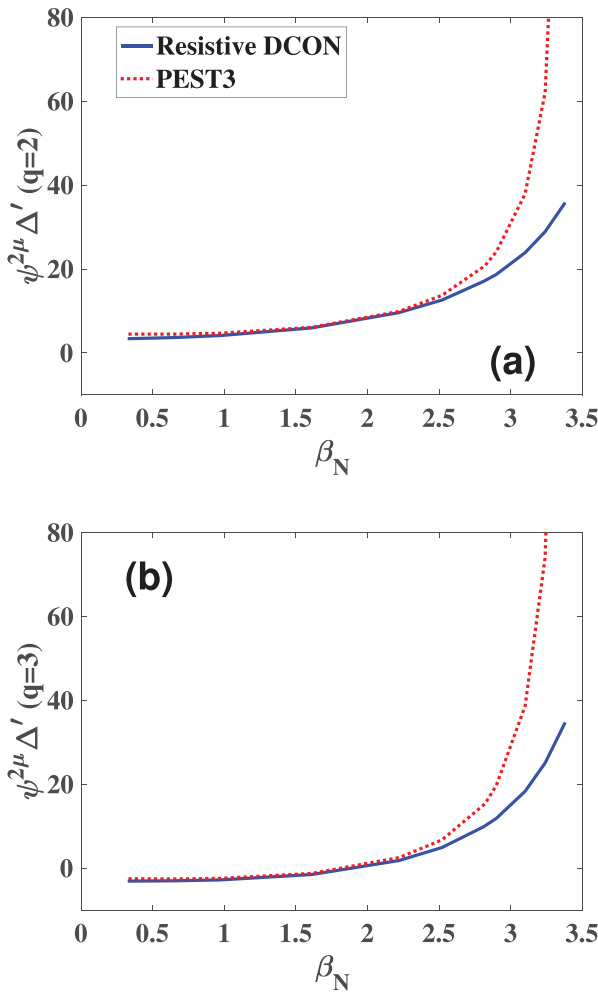


FIG. 6. Benchmark of Δ' at $q=2$ (a) and $q=3$ (b) resonant surfaces between the RDCON and PEST3 codes for $n=1$ mode.

this resistive response simulation. Figure 7 illustrates the resistive plasma response, which is driven by the edge perturbation with only $(m, n) = (2, 1)$ harmonic. Different from the plasma response in the cylindrical plasma, besides the $(2, 1)$ component, the other poloidal Fourier harmonics are also responding to the applied external perturbation due to the toroidal coupling effect of poloidal modes in the toroidal geometry. Additionally, each resonant harmonic at the corresponding resonant surface in Fig. 7 shows an odd parity of radial displacement, which indicates that the small tearing island can exist at each resonant surface in the toroidal case. Figure 8 illustrates the radial magnetic perturbations δB_r of the resistive plasma response by applying $(2, 1)$ harmonic dominated external perturbations. In Fig. 9, the resistive response is further compared with the ideal plasma response solved by the IPEC code. In general, the ideal and resistive response shows a similar structure, meanwhile the key difference appears at the resonant surface. In the straight field line coordinates, the ideal plasma response does not allow to open the island and keeps the resonant component of δB_r be zero at the corresponding resonant surface, since

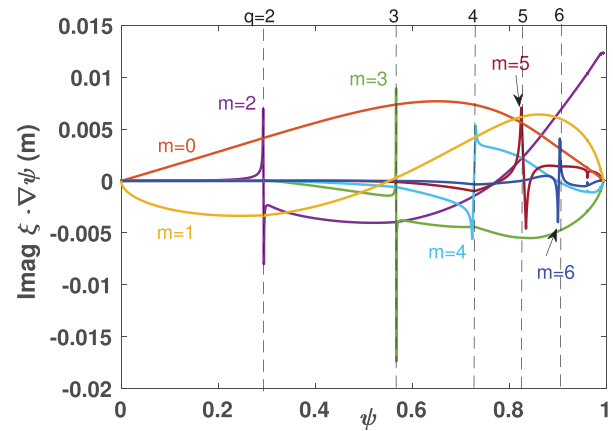


FIG. 7. The imaginary part (dominant part) of radial plasma displacement is plotted as a function of ψ . The major poloidal harmonics are plotted in the PEST coordinates.

$\delta B_r \sim [m - nq(\psi)]\xi_{\psi}^m$ in ideal MHD,⁴⁶ and $m - nq(\psi) \rightarrow 0$ when $\psi \rightarrow \psi_i^r$. However, in the case of the resistive plasma response, each resonant Fourier harmonic of δB_r in Fig. 9 keeps a finite value at the corresponding resonant surface due to the finite resistivity in Eq. (1b), which allows the existence of a small island. Figure 8 shows more clear behavior that the $m=3, 5$, and 6 poloidal harmonics of δB_r have finite values at $q=3, 5$, and 6 surfaces, respectively. In Fig. 9, it is noticed that the amplitude of radial magnetic perturbation shows a peak of $m=6$ harmonic near the $q=6$ resonant surface, which is just a local minimum point but not a singular point. The odd parity of radial displacement still appears at the $q=6$ surface in Fig. 7. Furthermore, Fig. 9 shows that the resistive response has stronger amplification of the magnetic response than the ideal response due to the field penetration in the presence of the opened island. This amplification has been observed in the KSTAR experiment. Ref. 39 reported that the plasma response can be greatly amplified by the magnetic island in the KSTAR experiment. The simulated resistive plasma response shows

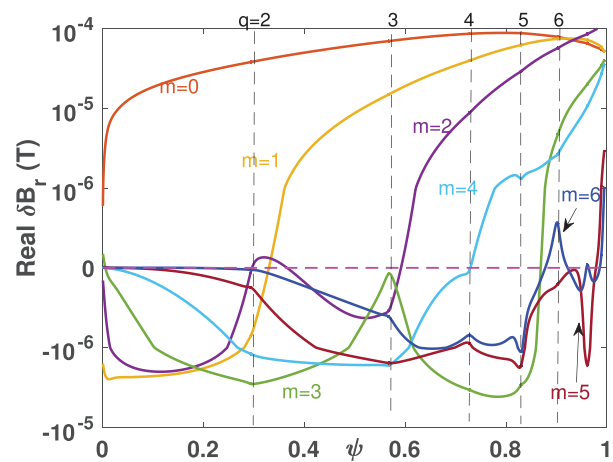


FIG. 8. The real part (dominant part) of magnetic perturbation is plotted as a function of ψ . The major poloidal harmonics are plotted in the PEST coordinates.

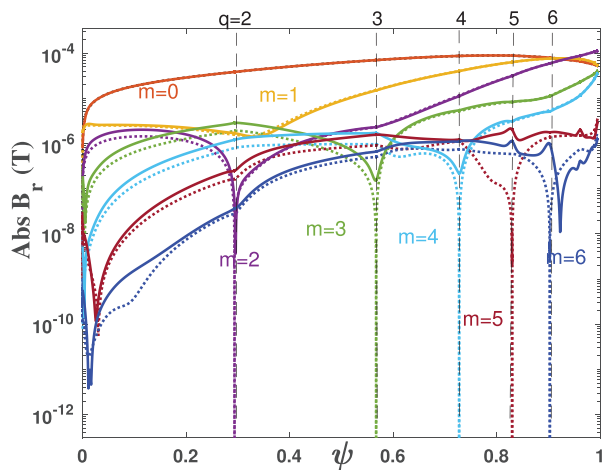


FIG. 9. The amplitude of radial magnetic perturbation is plotted as a function of ψ , where the resistive plasma response (solid line) and the ideal plasma response (dotted line) are compared. The PEST coordinates are adopted.

good agreement with the experimental observation, where the amplitude of the ideal response is much smaller. Reference 39 found that the resistive plasma response plays an essential role in the prediction of neoclassical toroidal viscosity (NTV) and finally revealed that the thermal electron contributed NTV torque in the KSTAR experiment. As illustrated in this section, the theoretical basis of toroidal asymptotic matching, developed in this work, can be a powerful tool to study and predict the resistive plasma response in tokamak experiments.

V. SUMMARY

This work makes progress on developing the method of solving the resistive plasma response based on asymptotic matching in the full toroidal geometry. The systematic benchmark of the RDCON code with analytical results and the PEST3 code confirms that the RDCON can solve the outer region reliably. This progress provides a solid base for applying the RDCON code in the analysis of tokamak experiments. In the development of solving the resistive plasma response, the derived matching matrix Eq. (11), determining the resistive plasma response, shows the way of 3D fields providing an additional small solution to interact with the resistive layers. It is different from the cylindrical geometry, where only one resonant surface is affected by the corresponding resonant harmonics of external 3D fields. In the full toroidal geometry, the small solutions, driven by the external magnetic perturbations, contribute the additional Δ at each resonant surface. The outer region Δ' matrix is challenging to solve in the toroidal geometry. The term, solved by the RDCON code, is carefully validated by reproducing the analytical Δ' in Ref. 64. The RDCON code also shows a good agreement of Δ' with the other region solver, PEST3 code. However, compared with PEST3, the lower Δ' value of the RDCON code while approaching the no-wall β limit still needs to be further understood. These numerical verifications confirm the reliability of the RDCON code as the outer region solver. It is also noted that increasing the plasma pressure can amplify Δ' value, which has the potential to drive the tearing instability. In contrast, the GGJ effect due to plasma pressure plays a stabilizing role in the inner region.

Therefore, instead of simply using the outer region Δ' to evaluate the stability of tearing mode, the result suggests that it is necessary to solve the eigenvalue problem of the matching matrix to determine tearing mode stability.⁴⁵ The resistive plasma response, simulated by the RDCON code, shows that there is a finite open island at multiple resonant surfaces due to the toroidal coupling effect of poloidal modes, though a single poloidal harmonic is applied. Meanwhile, compared with the ideal plasma response, the resistive layer can result in the amplification of the plasma response structure. At present, the code has been integrated with the GPEC project⁶⁸ to solve the resistive plasma response with many 3D coil geometries used in such as DIII-D, NSTX, and ITER tokamaks. The code will help to better understand the behavior of the plasma response by extending the resistive layer physics, such as the rotation effect, in the future application of experimental cases.

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DATA AVAILABILITY

The digital data for this paper can be found at Princeton Plasma Physics Laboratory <https://arks.princeton.edu/ark:/88435/dsp0141687m529>, Ref. 69.

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